Presentation 2 speaker notes

Vendor concentration in federal civilian contracting, FY2008 to FY2025. Suchitra Hole, Shivram Nekkanti and Andy Hannah.

Each page shows one slide, a short description of it, and the points to make. Say them in your own words. The same notes are in the PowerPoint file. Slides 1 to 15 are the talk. Slides 16 to 29 are the appendix for questions.

Slides

Slide	Title
1	Is federal contract money going to fewer companies?
2	In one VA market, a single firm gets 98% of the money
3	Six terms we use throughout
4	Our score: square each firm's share, then add them up
5	We built 365,000 markets from 88.8 million contract actions
6	How we tested each finding
7	Half of civilian contract dollars are highly concentrated
8	More markets rose than chance alone would produce
9	Across all 18 years, the rise beats chance, and VA drives it
10	Three VA health markets are each held by one firm
11	VA gave each region of its care network to one firm
12	Watch one-firm markets before they are competed again
13	Our firmest trend claim covers FY2012 to FY2020
14	Next: make the years since FY2020 comparable
15	Thank you
16	Appendix
17	Backup: one bad entry can fake a monopoly
18	Backup: the data decisions, in plain words
19	Backup: how we turned 18 zip files into a trusted dataset
20	Backup: the HHI for the real VA market, worked through
21	Backup: what counts as a rising market
22	Backup: rising beats chance clearly, falling only slightly
23	Backup: spreading contracts calms swings by a quarter
24	Backup: half the FY2021 jump came from new IT codes
25	Backup: the FY2024 dip was one firm split across four IDs
26	Backup: by dollars, VA accounts for the change
27	Backup: concentrated, but competed
28	Backup: Defense and civilian follow a similar path
29	Backup: civilian services are where concentration grew

Slide 1 · Talk

Is federal contract money going to fewer companies?



On the slide

Title, team and course.

Points to make

- Our question is whether federal contract money goes to fewer companies.
- We look at civilian agencies, every agency except Defense, from FY2008 to FY2025.
- Last time we showed the data and the measure. Today we show results.

Slide 2 · Talk

In one VA market, a single firm gets 98% of the money



On the slide

One bar splits VA's $22.8 billion managed health care market by firm. Three reasons it matters sit below.

Points to make

- In FY2025 VA spent $22.8 billion on managed health care contracts.
- This is mostly the networks of private doctors that treat veterans outside VA hospitals.
- UnitedHealth, through Optum, got 98%. 74 other firms shared 2%.
- One firm with most of a market means less price pressure, no quick fallback and fewer bidders next time.
- Concentration is not wrongdoing. Some work only one firm can do.
- Civilian agencies spent $301 billion on contracts in FY2025. Defense studies its own suppliers. We found no comparable measure for civilian agencies.
- Our question is how much civilian money sits in markets like this, and whether that is growing.

Slide 3 · Talk

Six terms we use throughout



On the slide

Six terms, each with a definition and an example in orange.

Points to make

- HHI is the standard concentration score the Justice Department and the FTC use for mergers.
- A market is one agency buying one kind of product in one year. Our files call it a segment.
- Above 2,500 is highly concentrated.
- Civilian means every agency except Defense.
- The fiscal year runs October 1 to September 30.
- Competed means other firms could bid. Sole source means the agency picked one firm.

Slide 4 · Talk

Our score: square each firm's share, then add them up

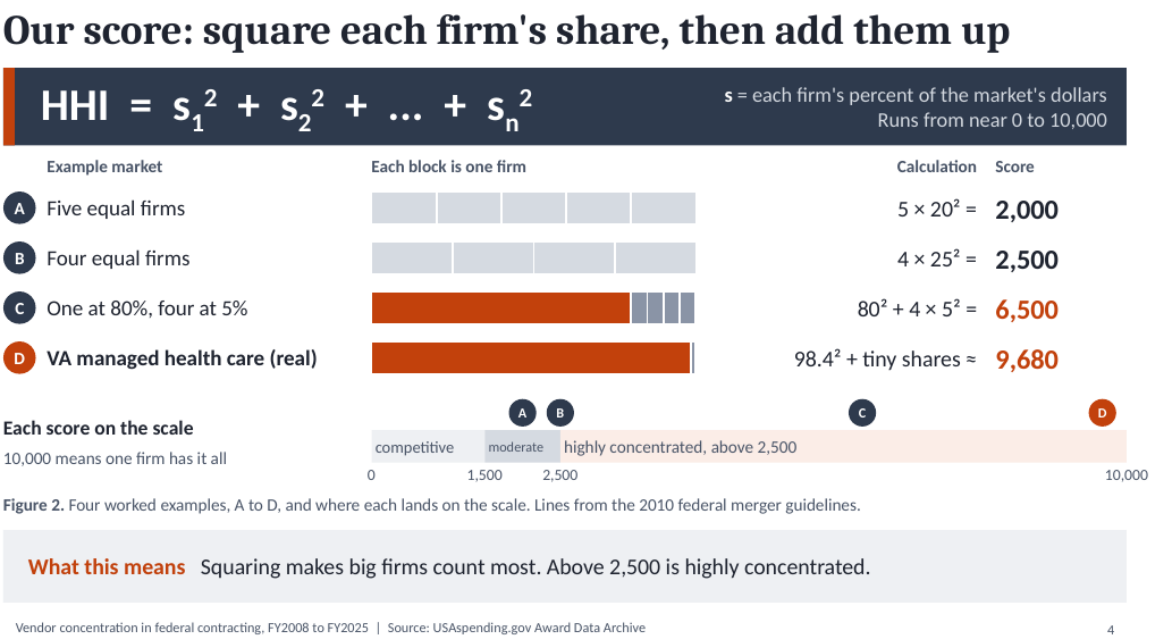


Figure 2. Four worked examples, A to D, and where each lands on the scale. Lines from the 2010 federal merger guidelines.

What this means Squaring makes big firms count most. Above 2,500 is highly concentrated.

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive

4

On the slide

The HHI formula, four worked examples and the 0 to 10,000 scale.

Points to make

- Take each firm's percent of the market, square it and add them up.
- Five equal firms score 2,000. Four equal firms score 2,500, right at the line.
- One firm at 80% scores 6,400 by itself. The four small firms add 100.
- Squaring makes big shares dominate the score.
- The real VA market scores 9,680. Its 74 small firms barely move it.
- We use 2,500 from the 2010 merger guidelines. The 2023 guidelines use 1,800, so anything above 2,500 meets both.

Slide 5 · Talk

We built 365,000 markets from 88.8 million contract actions



On the slide

One real contract action on the left. How actions group into a market on the right.

Points to make

- USAspending.gov records every federal contract action. One row is one action.
- The example is a real $857 million VA payment to UnitedHealth, fully competed.
- We use all 88.8 million rows from FY2008 to FY2025 and check each one against 62 rules.
- A market is one agency, one product and one year. That gives 365,000 markets, each with its own HHI.
- Firms count under their parent company.
- Our yearly totals match GAO's published figures within 0.2%.
- Cleaning matters. One bad entry made a market of 1,039 firms look like a monopoly, with an HHI of 4,282. Cleaned, it scores 167. The appendix has the full example.

Slide 6 · Talk

How we tested each finding



How we tested each finding

	The question	The method	In plain words
1	Is the concentrated share rising?	Yearly share	Add up the dollars in markets above 2,500 each year. Compare years with the same product codes.
2	Did single markets rise more than chance?	Shuffle test, market by market	Shuffle each market's yearly scores 20 times and count how many still rise.
3	Is the overall rise real, and is it VA?	Shuffle test on the yearly share	Shuffle the 18 yearly shares 20,000 times, with and without VA.
4	Is it just contract timing?	Spreading	Spread each contract's money evenly over the days it ran, then score again.
5	Why did VA concentrate?	Breakdown and case study	Split the change into each agency's part. Match VA's jump to the law and the contracts.

Table 1. The questions, the method behind each and what the method does.

What this means Each test asks whether the pattern is bigger than chance or one big contract would produce.

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive 6

On the slide

Five questions, the method behind each, and what the method does in plain words.

Points to make

- This is the map for the results that follow.
- The yearly share shows whether more money sits in concentrated markets.
- Two shuffle tests check that the rise is more than chance. One works market by market. The other works on the yearly share.
- Spreading checks whether contracts that pay several years at once cause the swings.
- The breakdown by agency and the VA case study show where the change came from.

Slide 7 · Talk

Half of civilian contract dollars are highly concentrated

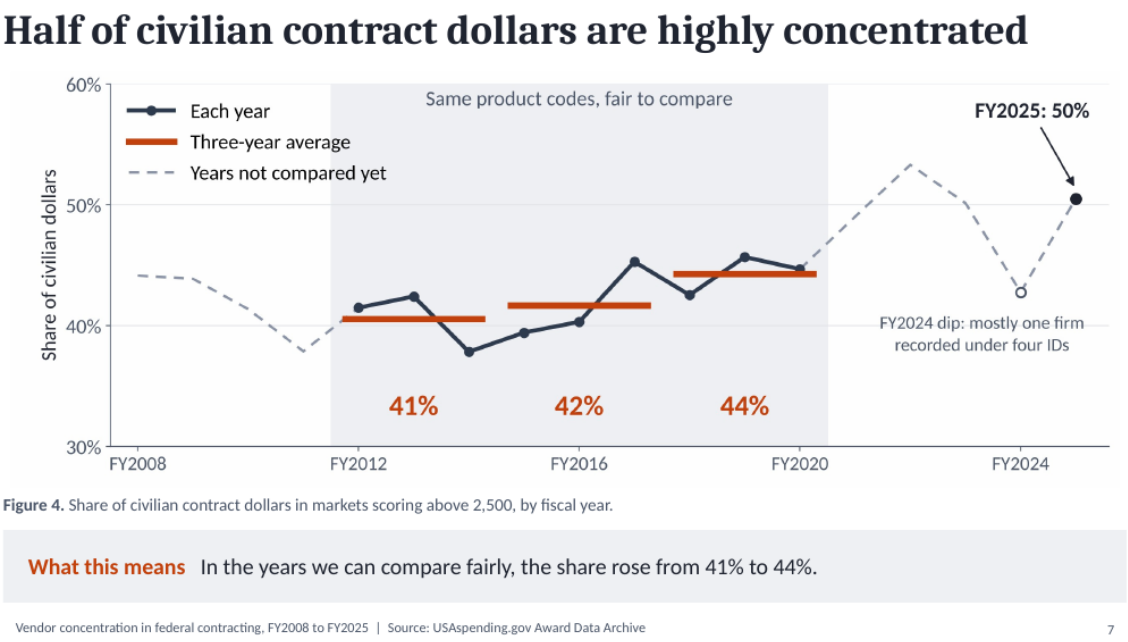


Figure 4. Share of civilian contract dollars in markets scoring above 2,500, by fiscal year.

What this means In the years we can compare fairly, the share rose from 41% to 44%.

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive 7

On the slide

The yearly share of civilian dollars in highly concentrated markets, with three-year averages in orange.

Points to make

- Each point is the share of civilian contract dollars in markets above 2,500 that year.
- In FY2025 it was about half.
- FY2012 to FY2020 use the same product codes, so those years compare fairly.
- Over those years the three-year average went from 41% to 42% to 44%.
- Four other ways of counting all end the period 3 to 6 points higher.
- VA drives much of it. Without VA the same averages go from about 41% to 42%.
- The FY2024 dip is mostly Honeywell recorded under four vendor IDs. Counted as one firm, the dip mostly closes.

Slide 8 · Talk

More markets rose than chance alone would produce

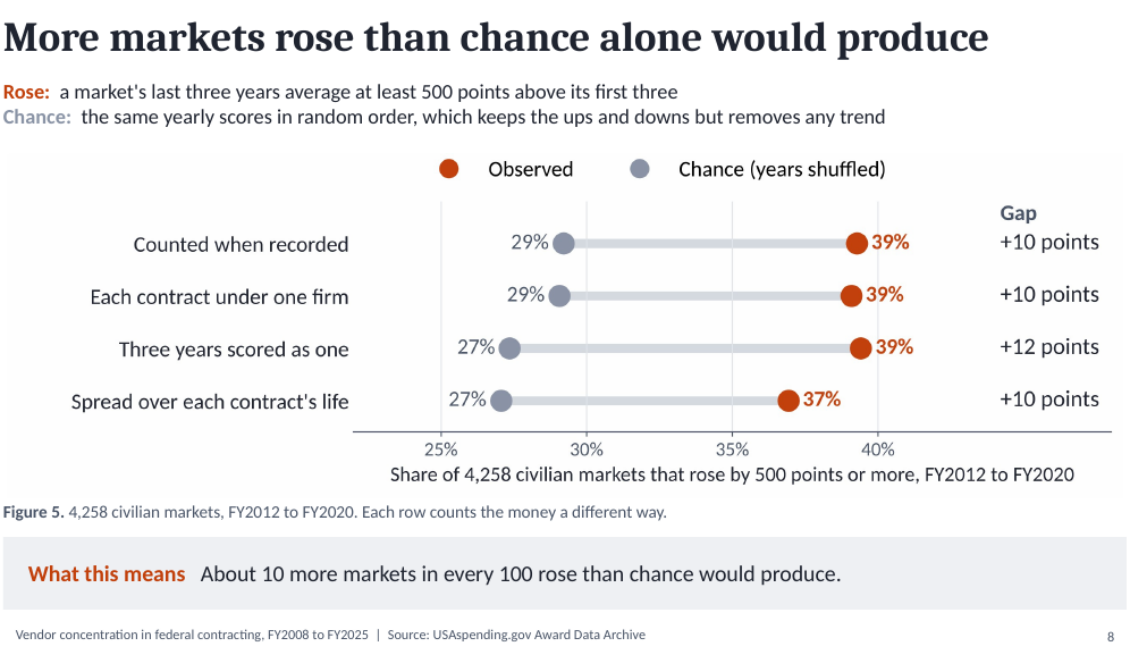


Figure 5. 4,258 civilian markets, FY2012 to FY2020. Each row counts the money a different way.

What this means About 10 more markets in every 100 rose than chance would produce.

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive 8

On the slide

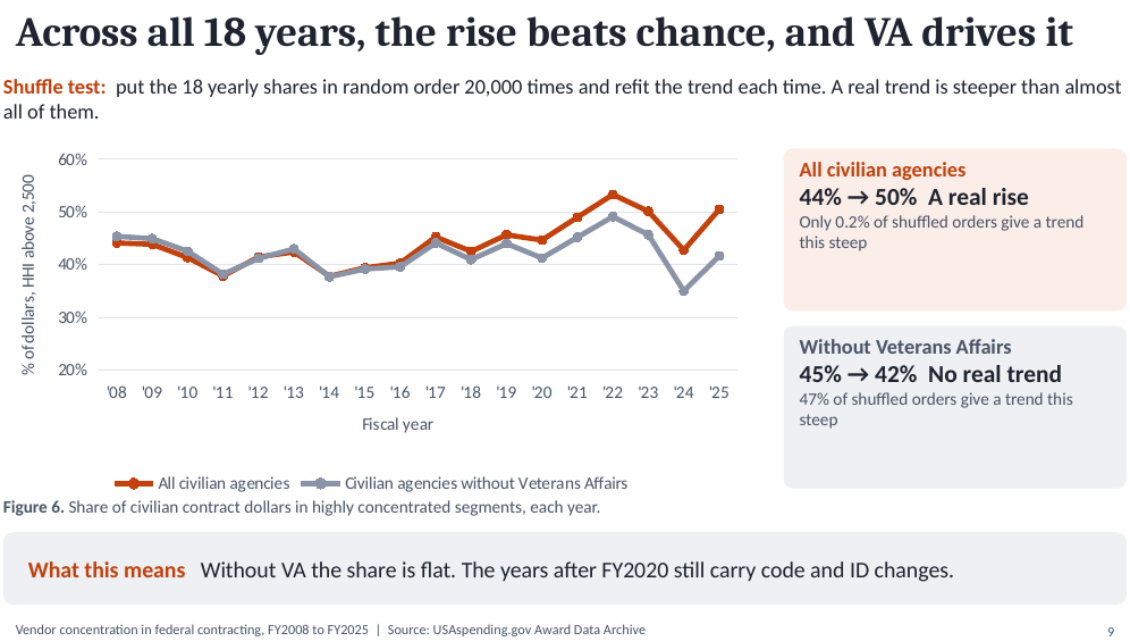
Four ways of counting the money. Each row compares civilian markets that rose with markets that rose by chance.

Points to make

- We follow each civilian market from FY2012 to FY2020. There are 4,258 of them.
- A market rose if its last three years average at least 500 points above its first three.
- Chance means shuffling each market's yearly scores into random order, 20 times. That keeps the swings and removes any trend.
- Counted as recorded, 39% of civilian markets rose. Chance gives 29%.
- The gap stays between 10 and 12 points every way we count. It also holds by dollars.
- Spreading contracts over their lives removes about a quarter of the swings. The gap still holds.
- Defense on its own shows a smaller gap, about 8 to 9 points.

Slide 9 · Talk

Across all 18 years, the rise beats chance, and VA drives it



On the slide

The civilian share each year with and without VA, and the result of an 18-year shuffle test for each.

Points to make

- This is a second test of the rise, on the yearly share of dollars across all 18 years.
- We shuffled the 18 yearly values 20,000 times and refit the trend each time.
- All civilian agencies go from 44% to 50%. Only about 0.2% of shuffles give a trend this steep.
- Without VA the share goes from 45% to 42%. 47% of shuffles are as steep, so there is no real trend.
- Both tests say the rise is more than chance. This one adds that VA drives it.
- This test runs across the FY2021 code change and the FY2024 ID split. The years after FY2020 are provisional.

Slide 10 · Talk

Three VA health markets are each held by one firm

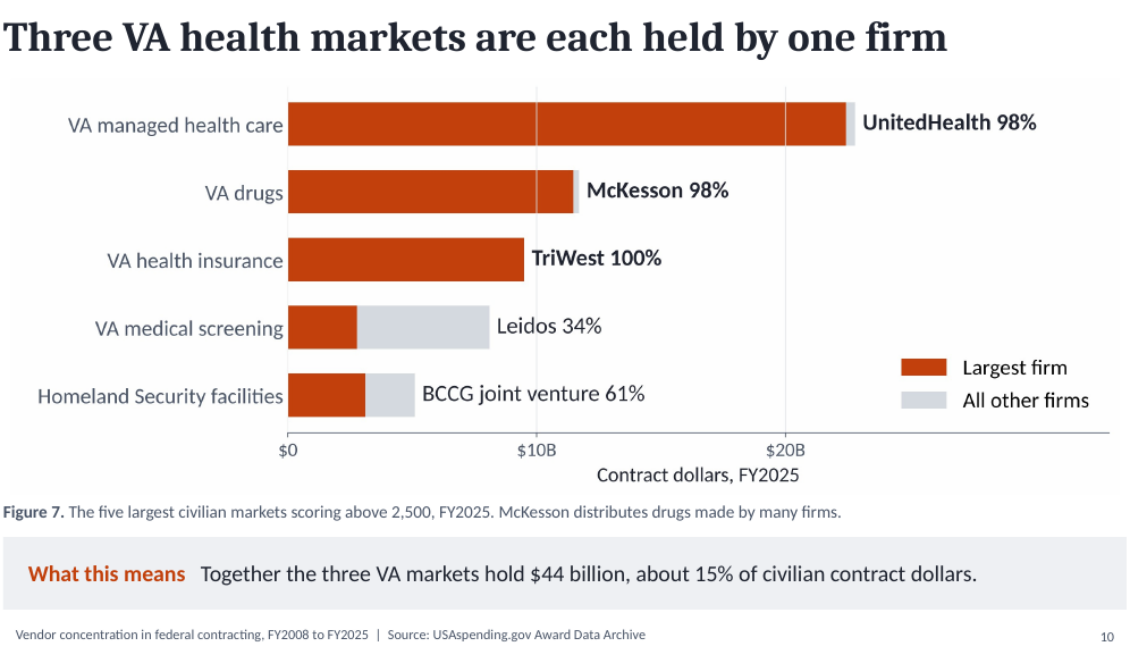


Figure 7. The five largest civilian markets scoring above 2,500, FY2025. McKesson distributes drugs made by many firms.

What this means Together the three VA markets hold $44 billion, about 15% of civilian contract dollars.

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive

10

On the slide

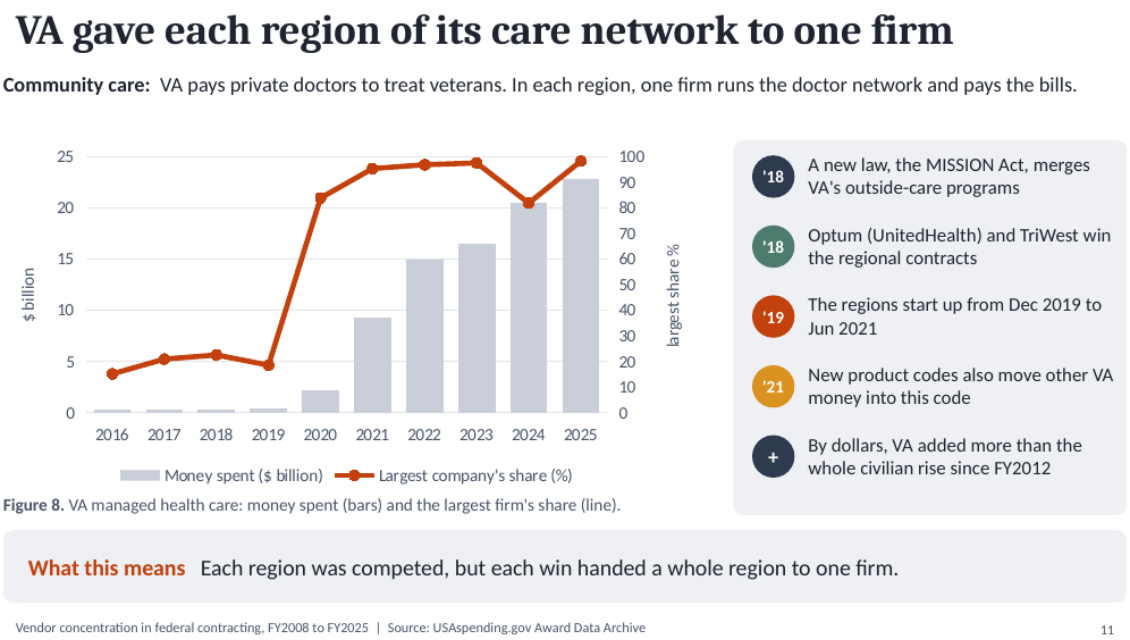
The five largest highly concentrated civilian markets in FY2025, with the largest firm's share in orange.

Points to make

- Three of the five are VA health markets.
- UnitedHealth has 98% of managed health care. It runs most of VA's community care network.
- McKesson has 98% of VA drugs. It is the distributor, and the drugs come from many makers.
- TriWest has essentially all of VA health insurance.
- Together the three hold $44 billion, about 15% of civilian contract dollars.
- Leidos leads VA medical screening with 34%. A few other large firms hold most of the rest.

Slide 11 · Talk

VA gave each region of its care network to one firm



On the slide

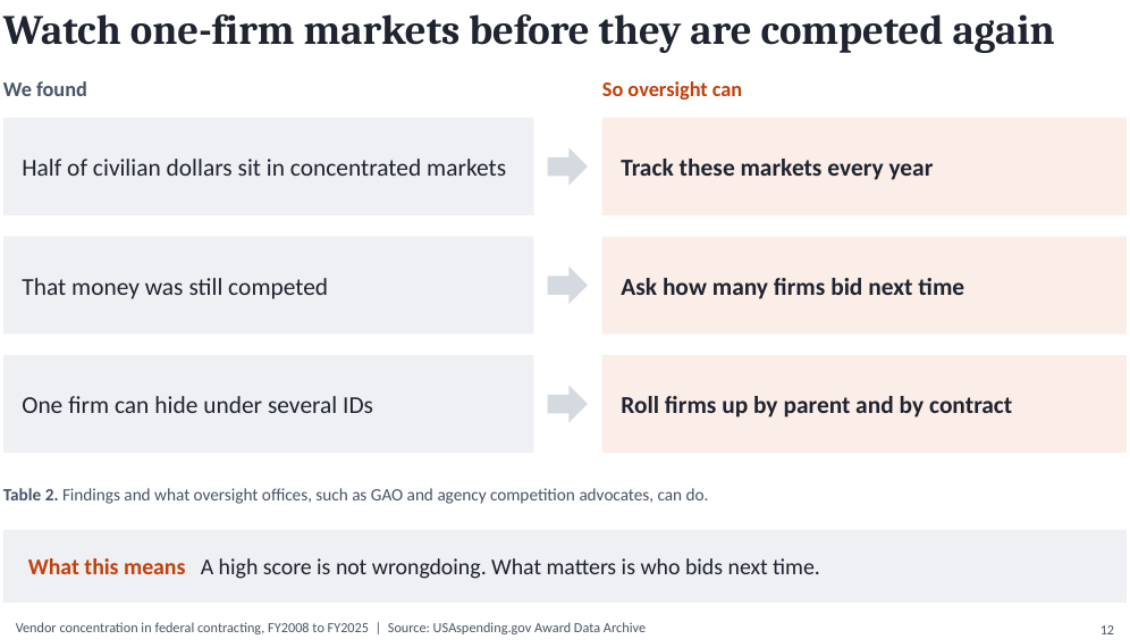
VA managed health care dollars by year as bars, the largest firm's share as a line, and four dated events.

Points to make

- Community care is when VA pays private doctors to treat veterans.
- In each region one firm builds the doctor network and pays the bills.
- The MISSION Act in June 2018 merged VA's outside-care programs.
- VA competed the regional contracts. Optum, part of UnitedHealth, and TriWest won them.
- The regions started up from December 2019 to June 2021.
- In 2019 the code held about $400 million and the largest firm had 19%. By 2025 it held $22.8 billion and UnitedHealth had 98%.
- New product codes in FY2021 also moved other VA money into this code. Part of the jump is relabeling.
- 99.6% to 100% of these dollars are marked competed. Each win covered a whole region.
- Comparing FY2012 to FY2014 with FY2023 to FY2025, VA added 11.6 points. The whole civilian rise was 7.2 points.

Slide 12 · Talk

Watch one-firm markets before they are competed again



On the slide

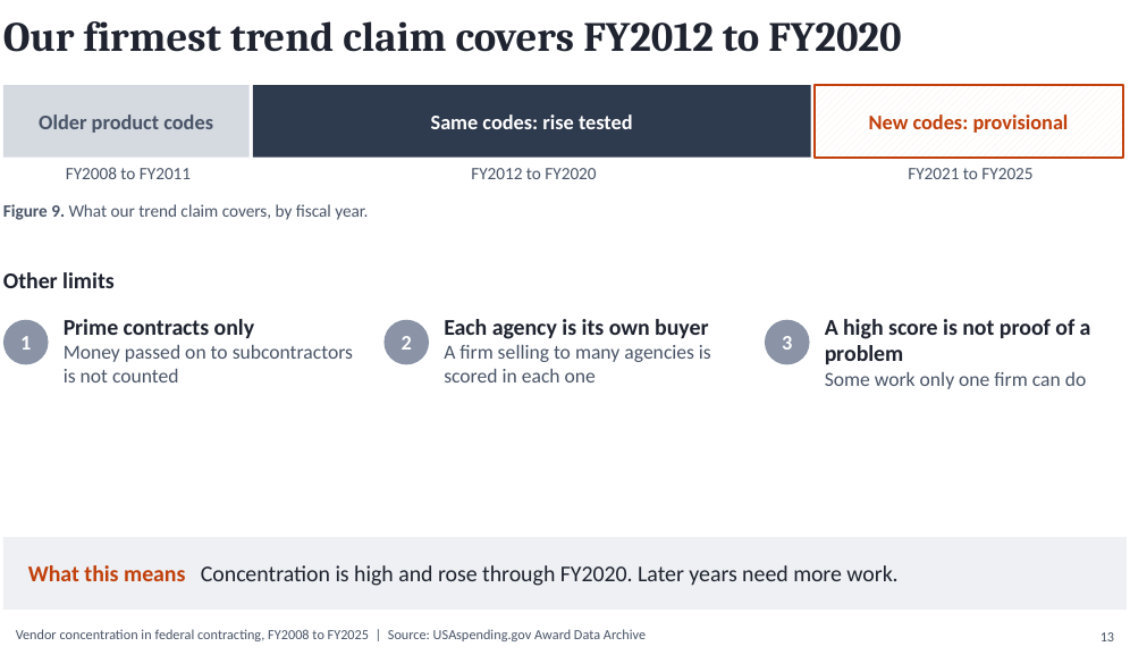
Three findings on the left and what oversight offices can do on the right.

Points to make

- Half of civilian dollars sit in concentrated markets. GAO and agency competition advocates could track these markets every year.
- The money was still competed. 85% of concentrated dollars against 86% of other dollars, among dollars with a competition code.
- Competition happens once for a whole market. The winner then holds it for years.
- The moment that matters is the next competition and how many firms bid.
- One firm can show up under several IDs. Roll firms up by parent and by contract.
- A high score is not wrongdoing.

Slide 13 · Talk

Our firmest trend claim covers FY2012 to FY2020



On the slide

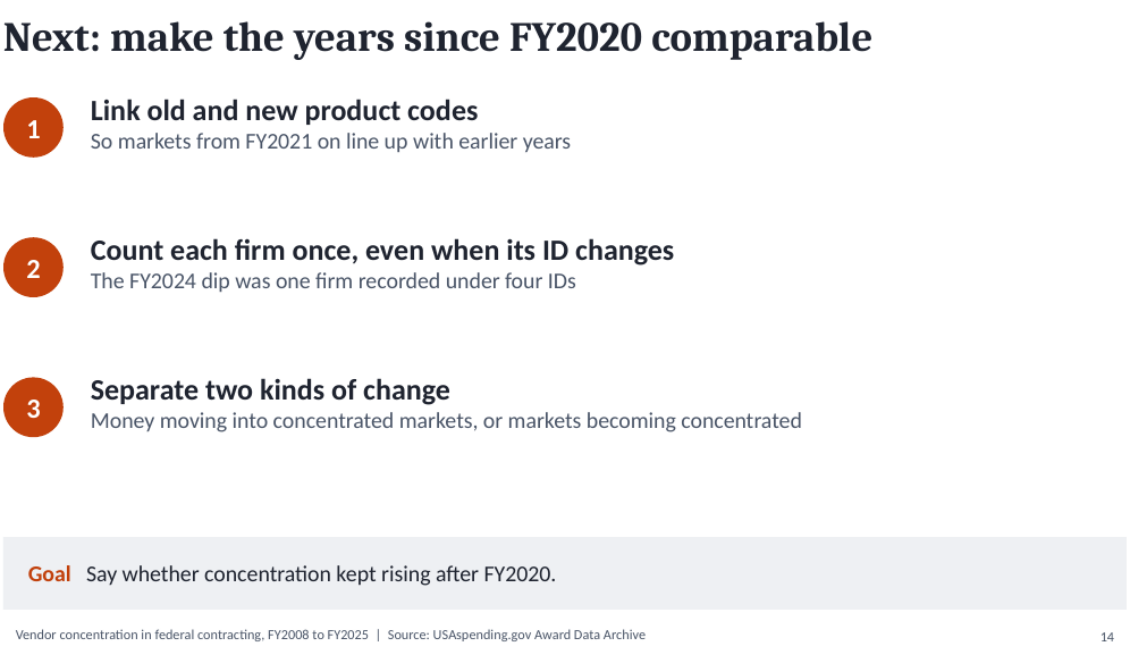
Which years our firmest trend claim covers, and three other limits.

Points to make

- Our firmest trend claim covers FY2012 to FY2020, where the product codes stay the same.
- The share since FY2020 is higher, and the 18-year test agrees. The codes changed in FY2021 and some firms changed IDs, so those years are provisional.
- Our data covers prime contracts only. Money passed on to subcontractors is not counted.
- Each agency is its own buyer. A version that pools all civilian agencies still rose.
- A high score is not proof of a problem.

Slide 14 · Talk

Next: make the years since FY2020 comparable



On the slide

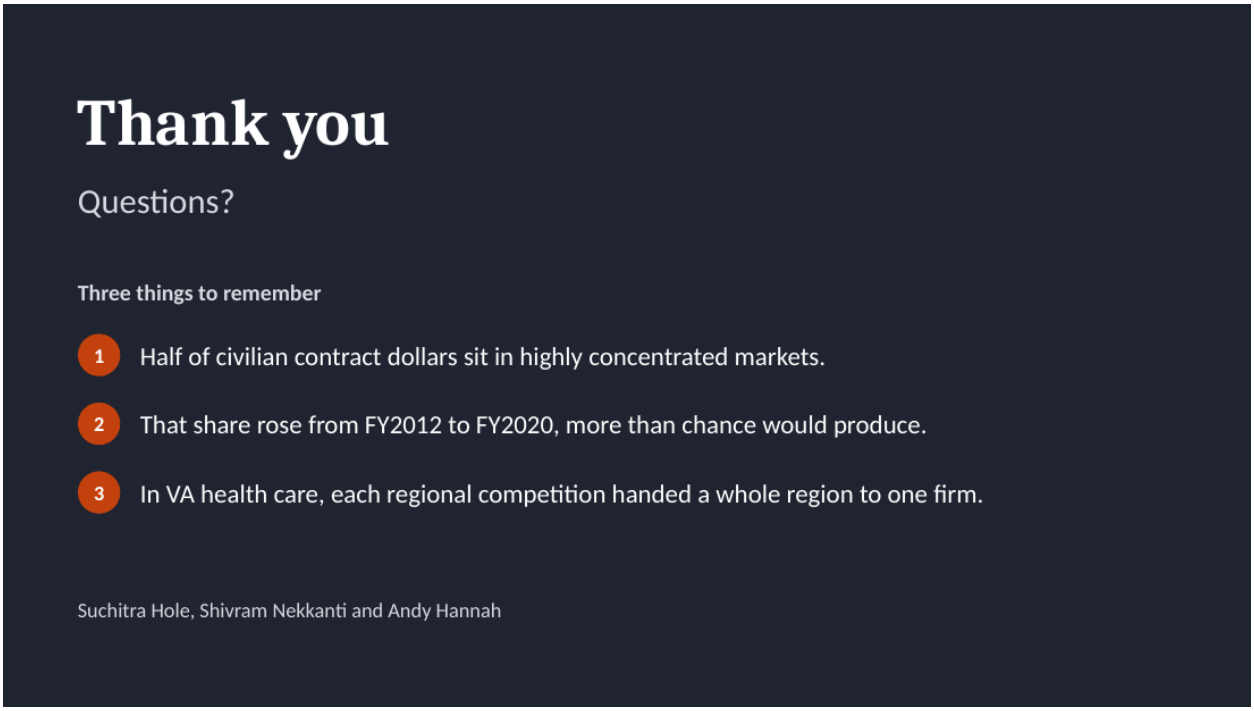
Three next steps, each with its reason.

Points to make

- Link old and new product codes. About half of the FY2021 jump comes from IT codes splitting into finer ones.
- Count each firm once, even when its ID changes. The FY2024 dip was one firm under four IDs.
- Separate money moving toward concentrated markets from markets becoming concentrated. They mean different things for policy.
- With these done we can say whether concentration kept rising after FY2020.

Slide 15 · Talk

Thank you



On the slide

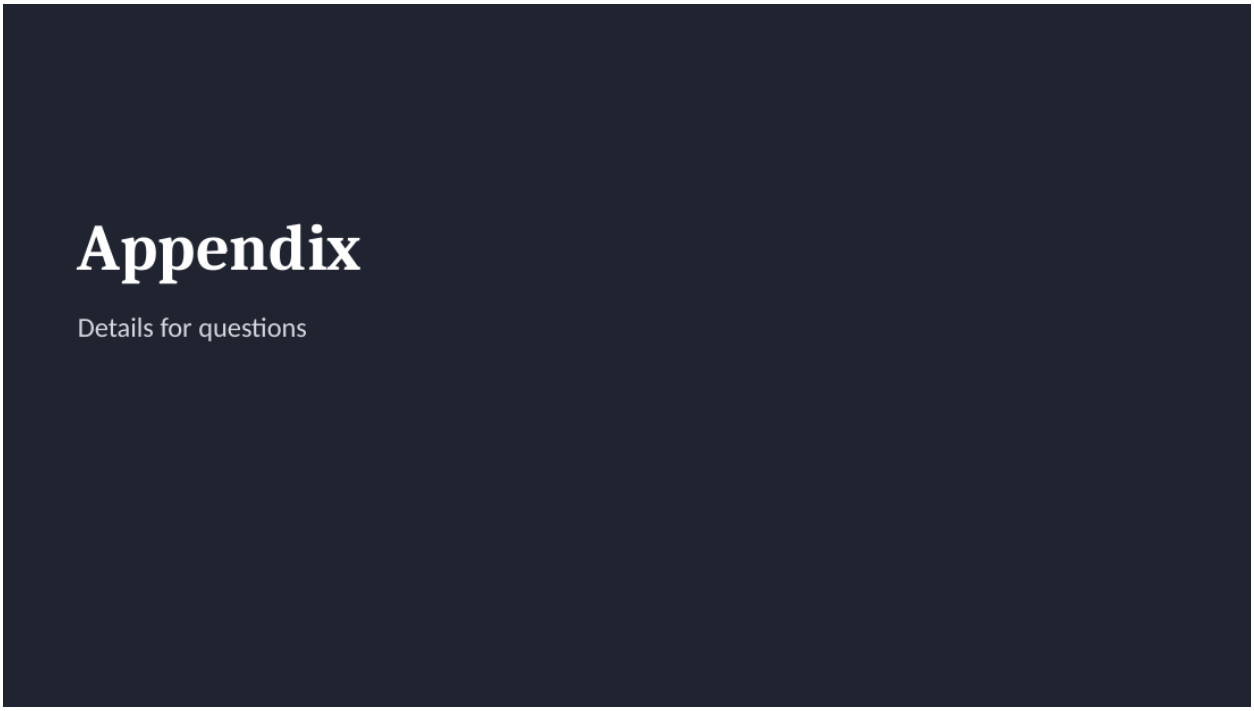
Thank you, questions, and three things to remember. Leave it up during questions.

Points to make

- Half of civilian contract dollars sit in highly concentrated markets.
- The share rose from FY2012 to FY2020, more than chance would produce.
- VA health care shows how one competition can hand a whole region to one firm.
- The appendix covers likely questions.

Slide 16 · Appendix

Appendix



On the slide

Appendix divider.

Points to make

- Slides 17 to 29 answer likely questions.

Slide 17 · Appendix

Backup: one bad entry can fake a monopoly



On the slide

Emerson's share of one Army construction market, and the market's HHI, in the raw data and after cleaning.

Question it answers

Why does cleaning matter?

Points to make

- In FY2008 the Army recorded a $13.9 billion entry for Emerson Construction and reversed it the next year.
- The real contract was about $16 million.
- With the entry left in, Emerson holds 65% of a market of 1,039 firms. The market scores 4,282.
- Netted against its reversal, Emerson holds 0.2%. The market scores 167.
- This one is a Defense market. The same checks run on every row.
- We found five issues like this. The next slide lists each one and the rule we propose.

Slide 18 · Appendix

Backup: the data decisions, in plain words

Backup: the data decisions, in plain words

The issue	What we do (as proposed)	Why
A $13.9B entry reversed the next year, and four similar pairs	Replace each pair with its real net	Otherwise one company looks dominant
Two $10B US Mint rows	Leave out of the score until confirmed	They look like contract ceilings, which are spending limits
Money returned makes some company totals negative	Score only companies with a positive total	A negative share has no meaning
One name, MISCELLANEOUS FOREIGN AWARDEES, covers many firms	Leave out of the score, report its share	It is not one company
Rows with no company ID	Leave out of the score, report their share	Under 0.01% of dollars
A placeholder funding agency in FY2008 to FY2010	Use the buying (awarding) agency	Complete in every year

Table B2. The proposed data decisions. With these rules, 3,241 of 357,284 markets (0.9%) change concentration band. The team still has to confirm each rule.

On the slide

A table of the six proposed data decisions, with what we do and why.

Question it answers

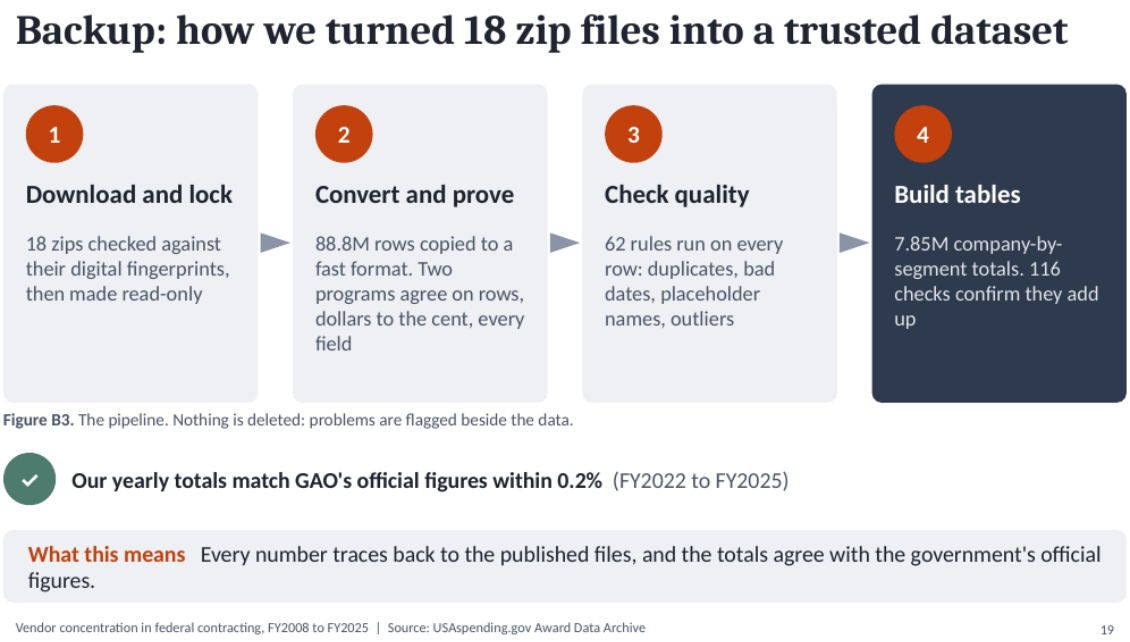
How did we clean the data?

Points to make

- Each row is one issue, what we do about it and why.
- Together the rules move 0.9% of markets across a concentration band.
- The team still has to confirm each rule.

Slide 19 · Appendix

Backup: how we turned 18 zip files into a trusted dataset



On the slide

The four-step data pipeline and the GAO check.

Question it answers

How do we know the data is right?

Points to make

- We checked all 18 files against their digital fingerprints and locked them.
- Two separate programs agree on row counts, dollars to the cent and every field.
- 62 quality rules run on every row.
- Our yearly totals match GAO within 0.2% for FY2022 to FY2025.

Slide 20 · Appendix

Backup: the HHI for the real VA market, worked through

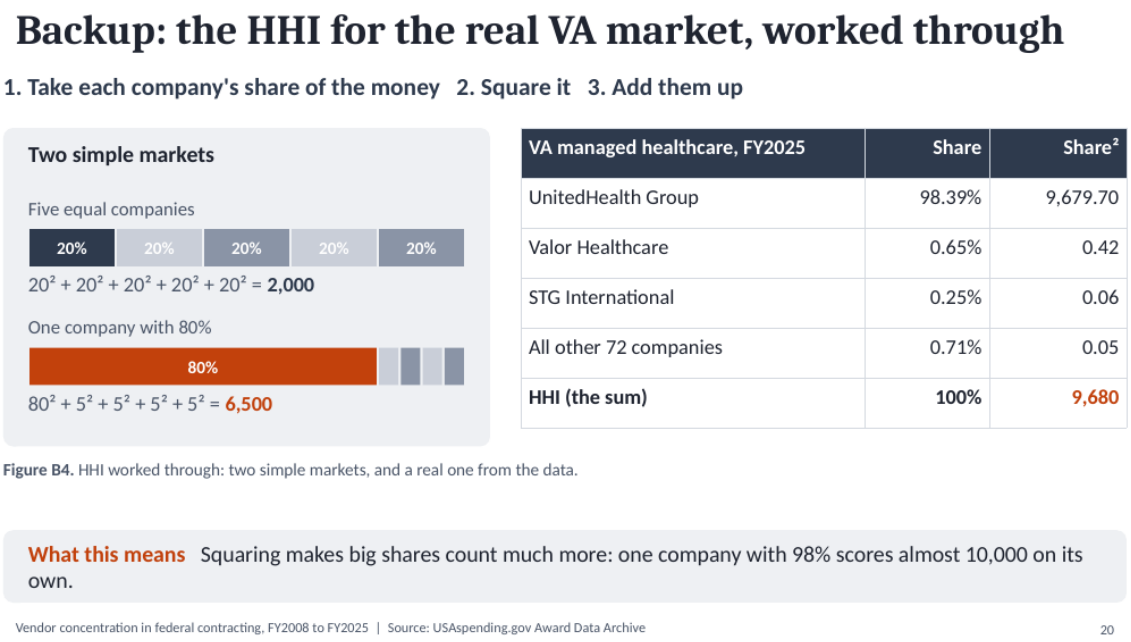


Figure B4. HHI worked through: two simple markets, and a real one from the data.

On the slide

The HHI calculation for the real VA market, firm by firm, next to two simple markets.

Question it answers

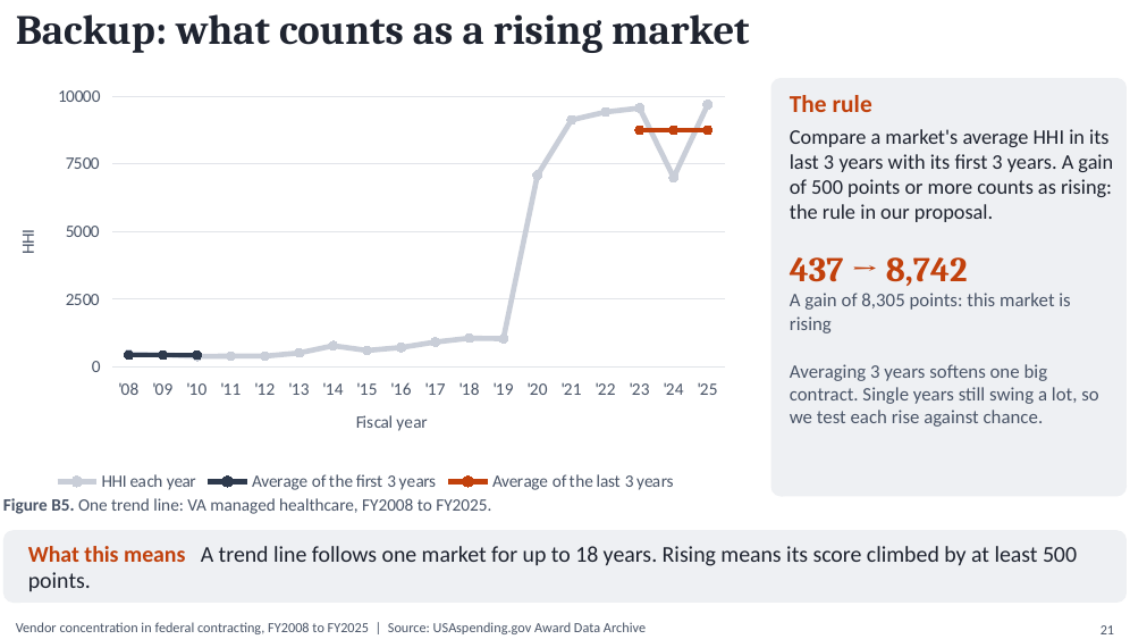
How exactly do you get 9,680?

Points to make

- UnitedHealth has 98.39%. Squared, that is 9,680 by itself.
- The next two firms add less than half a point.
- The other 72 firms together add 0.05.

Slide 21 · Appendix

Backup: what counts as a rising market



On the slide

One market's yearly HHI from FY2008 to FY2025, with its first and last three-year averages.

Question it answers

What counts as a rising market?

Points to make

- Compare the average of a market's last three years with its first three.
- A gain of 500 points or more counts as rising.
- VA managed health care went from 437 to 8,742.
- Averaging three years softens one big contract. Single years still swing a lot.

Slide 22 · Appendix

Backup: rising beats chance clearly, falling only slightly

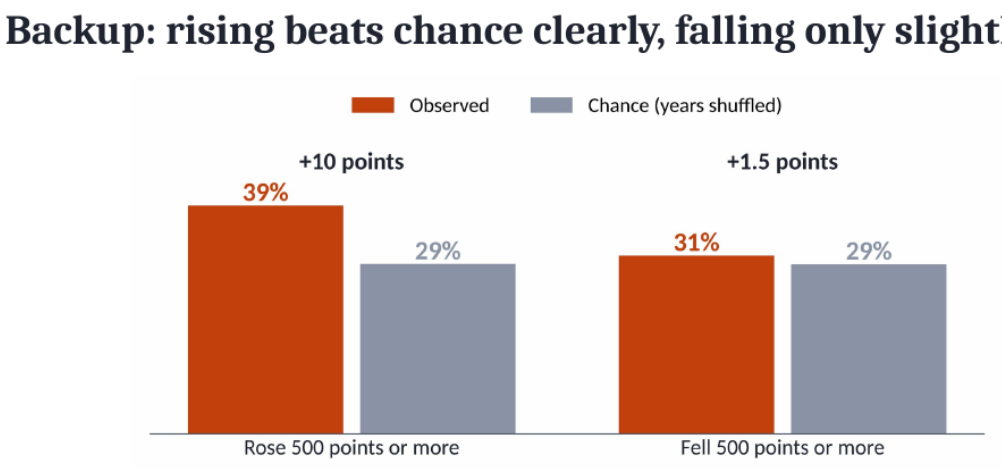


Figure B6. Share of 4,258 civilian markets that rose or fell by 500 points or more, FY2012 to FY2020, as recorded.

On the slide

Bars for civilian markets that rose and fell by 500 points or more, observed against chance.

Question it answers

Did some markets become less concentrated?

Points to make

- Counted as recorded, 39% of civilian markets rose against 29% by chance.
- 31% fell against 29% by chance.
- With contracts spread evenly over their lives, falling beats chance by about 4 points and rising by about 10.
- Markets move both ways. More of them rise.

Slide 23 · Appendix

Backup: spreading contracts calms swings by a quarter

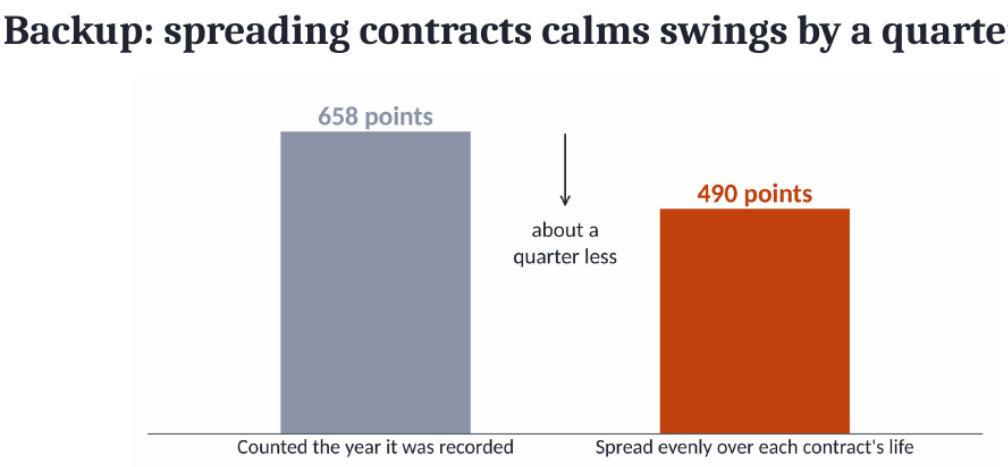


Figure B7. Typical yearly change in a civilian market's HHI, FY2012 to FY2020. Each contract counts under one firm.

On the slide

The typical yearly change in a market's HHI, as recorded and with contracts spread over their lives.

Question it answers

Are the swings just contract timing?

Points to make

- Some contracts record several years of money at once.
- We spread each contract's total evenly over the days it ran.
- The typical yearly change falls from 658 points to 490, about a quarter.
- Most of the swing remains. The rise still beats chance.
- We can only spread contracts that start and end inside our data.

Slide 24 · Appendix

Backup: half the FY2021 jump came from new IT codes

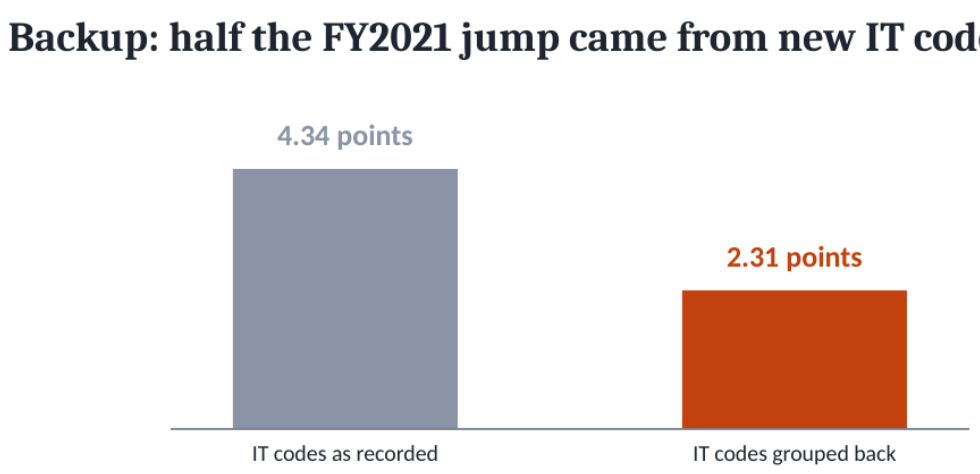


Figure B8. Rise in the civilian share above 2,500 from FY2020 to FY2021.

On the slide

The FY2020 to FY2021 rise in the civilian share, with IT codes as recorded and grouped back.

Question it answers

Why the jump in FY2021?

Points to make

- The FY2021 code manual split broad IT codes into many finer ones.
- Splitting a market makes each piece look more concentrated, even with the same firms.
- Grouping IT back into one services and one products market per agency cuts the jump from 4.34 points to 2.31.
- That is why the trend claim stops at FY2020 for now.

Slide 25 · Appendix

Backup: the FY2024 dip was one firm split across four IDs

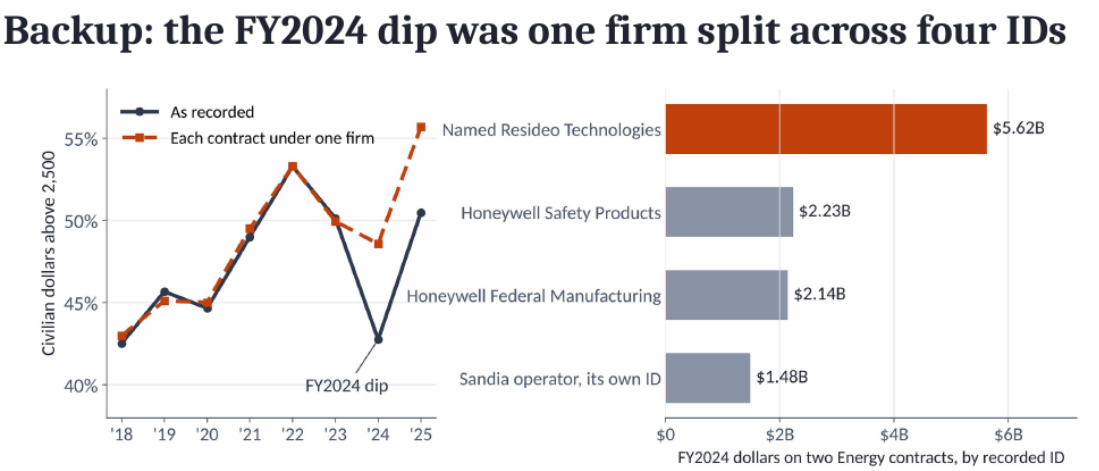


Figure B9. Left: civilian share above 2,500, two ways. Right: Honeywell's FY2024 money on two Energy contracts, by ID.

On the slide

The civilian share from FY2018 to FY2025 two ways, and Honeywell's FY2024 money by recorded ID.

Question it answers

What caused the FY2024 dip?

Points to make

- Honeywell runs Sandia National Laboratories and the Kansas City site for Energy.
- In FY2024 the parent field changes partway through the year. The same contracts show up under four IDs.
- One of them is named Resideo, a company Honeywell spun off in 2018.
- Split four ways, the market falls below 2,500.
- This one market gives 5.41 of the 5.82 point dip. Nothing changed in who runs those sites.

Slide 26 · Appendix

Backup: by dollars, VA accounts for the change

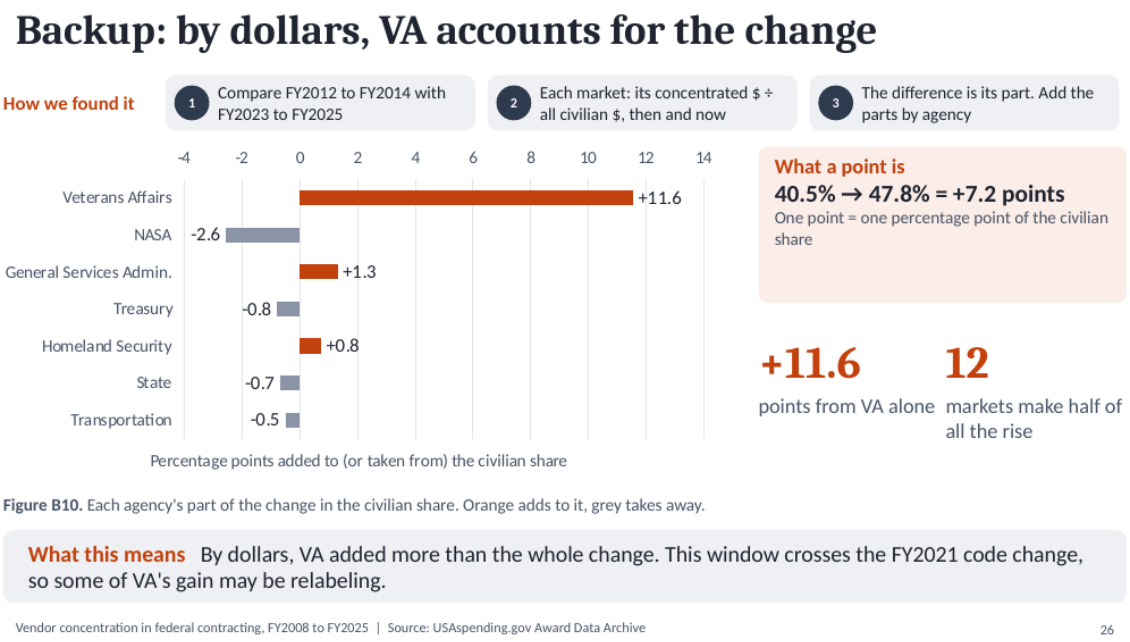


Figure B10. Each agency's part of the change in the civilian share. Orange adds to it, grey takes away.

What this means By dollars, VA added more than the whole change. This window crosses the FY2021 code change, so some of VA's gain may be relabeling.

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive 26

On the slide

Each agency's part of the change in the civilian share between FY2012 to FY2014 and FY2023 to FY2025.

Question it answers

Which agencies drove the change?

Points to make

- The share went from 40.5% to 47.8%, up 7.2 points.
- VA alone added 11.6 points, more than the whole change.
- Other agencies, led by NASA, took some away.
- The two windows sit on either side of the FY2021 code change. Some of VA's gain may be relabeling.

Slide 27 · Appendix

Backup: concentrated, but competed



Figure B11. How civilian contract dollars were awarded, FY2025.

On the slide

How FY2025 civilian dollars were awarded, in concentrated markets and in all others.

Question it answers

Is concentrated money simply not competed?

Points to make

- In FY2025, 90% of dollars in concentrated markets were competed. All other markets were also about 90%.
- Over all 18 years it is 85% against 86%.
- NASA is the exception, with about 37% of its concentrated dollars sole source, mostly space research.
- Orders under umbrella contracts may carry the umbrella's competition code.

Slide 28 · Appendix

Backup: Defense and civilian follow a similar path

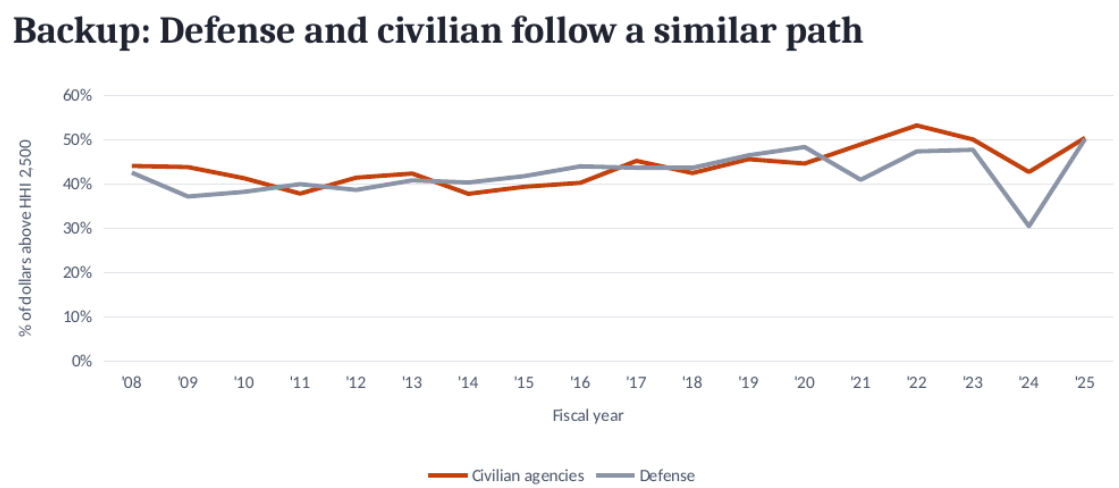


Figure B12. Share of dollars in highly concentrated segments, Defense and civilian.

On the slide

The share of dollars in highly concentrated markets each year, civilian against Defense.

Question it answers

What about Defense?

Points to make

- Defense follows a similar path.
- Defense rose more over FY2012 to FY2020, from about 40% to 46% by three-year averages.
- The Defense FY2024 dip comes from combat ships and fixed-wing aircraft. They come back in FY2025.
- We have not checked whether ID splits explain part of the Defense dip.

Slide 29 · Appendix

Backup: civilian services are where concentration grew

Backup: civilian services are where concentration grew

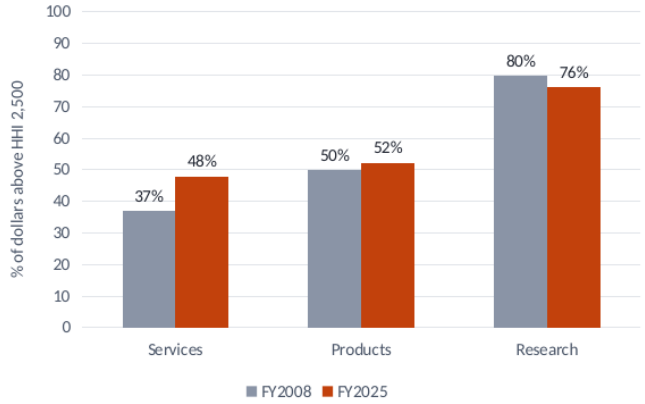


Figure B13. Civilian dollars in highly concentrated segments, by kind of purchase. Right: total civilian dollars by kind.

Civilian dollars	FY2008	FY2025
Services	$96.9B	$229B
Products	$31.3B	$52.1B
Research	$15.0B	$19.9B

Vendor concentration in federal contracting, FY2008 to FY2025 | Source: USAspending.gov Award Data Archive

On the slide

The concentrated share by kind of purchase in FY2008 and FY2025, with total dollars.

Question it answers

Which kinds of purchase grew more concentrated?

Points to make

- Services went from 37% to 48% concentrated while service spending more than doubled.
- Research stays highly concentrated throughout.
- Products barely moved.